Practical No : 2

Title: Email Spam Filtering Using Text Classification Algorithm

Aim:

The objective of this project is to design and implement an email spam filtering system using a text classification algorithm. The system will classify incoming emails as either "Spam" or "Not Spam" based on their content. The implementation will involve preprocessing the email text, applying a machine learning algorithm, and evaluating its performance using an appropriate dataset.

Prerequisite:

1. Knowledge and Skills:

- **Understanding of Natural Language Processing (NLP):** Familiarity with concepts such as text preprocessing, tokenization, stop word removal, stemming, and vectorization.
- **Machine Learning Fundamentals:** Basic understanding of supervised learning, classification algorithms (e.g., Naive Bayes, Support Vector Machines, or Logistic Regression), and evaluation metrics.
- **Programming Skills:** Proficiency in Python and experience with relevant libraries such as scikit-learn, NLTK, pandas, and NumPy.
- **Dataset Handling:** Ability to work with datasets, including loading, cleaning, and splitting data into training and test sets.

2. Tools and Libraries:

- **Python:** Programming language used for the implementation.
- **scikit-learn:** Machine learning library for implementing text classification algorithms.
- **NLTK or SpaCy:** For text preprocessing tasks.
- **pandas:** For data manipulation and analysis.
- **Jupyter Notebook:** IDE for writing and running Python code.

Input:

- **Dataset:** A labeled dataset of emails containing both "Spam" and "Not Spam" (Ham) examples. A commonly used dataset for this purpose is the **Enron Email Dataset** or the **SpamAssassin Public Corpus**.
    - **Features:** Email content (text data).
    - **Labels:** Spam (1) or Not Spam (0).

Sample Input:

- **Email Content:** "Congratulations! You've won a free vacation. Click here to claim your prize."
- **Label:** Spam (1)

Outcome:

- The primary outcome of this project is an email spam filter that can accurately classify emails as "Spam" or "Not Spam."
- **Performance Metrics:** The performance of the spam filter will be evaluated using accuracy, precision, recall, and F1-score.

Expected Outcome:

- **Input:** "Hello, your invoice for the recent purchase is attached."
- **Predicted Label:** Not Spam (0)

Theory:

1. Text Classification in NLP:

- **Text classification** is a supervised machine learning technique that assigns a label to a given piece of text. In the context of spam filtering, the text is the content of an email, and the label is whether the email is spam or not.

2. Preprocessing Steps:

- **Tokenization:** Breaking down the email content into individual words or tokens.
- **Stop Word Removal:** Eliminating common words that do not contribute significant meaning to the classification task.
- **Stemming/Lemmatization:** Reducing words to their root or base form to treat different forms of a word as the same feature.
- **Vectorization:** Converting the text data into numerical form using techniques like Bag of Words (BoW) or Term Frequency-Inverse Document Frequency (TF-IDF).

3. Classification Algorithm:

- **Naive Bayes Classifier:** A probabilistic classifier based on Bayes' theorem, often used in text classification due to its simplicity and effectiveness.
- **Support Vector Machine (SVM):** A powerful classifier that works by finding the hyperplane that best separates the data into different classes.
- **Logistic Regression:** A linear model used for binary classification tasks, predicting the probability that a given input belongs to a particular class.

4. Model Training and Evaluation:

- The dataset is split into training and testing sets. The model is trained on the training data and evaluated on the testing data.
- **Evaluation Metrics:**
    - **Accuracy:** The percentage of correctly classified emails.
    - **Precision:** The ratio of correctly predicted spam emails to the total predicted spam emails.
    - **Recall:** The ratio of correctly predicted spam emails to the total actual spam emails.
    - **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure of the classifier's performance.

Conclusion:

The implementation of an email spam filtering system using a text classification algorithm demonstrates the practical application of NLP and machine learning techniques in real-world problems. The project successfully filters spam emails with a high degree of accuracy, contributing to more efficient email management.